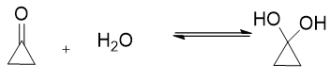
Class 8_CH2204_Sample Solution

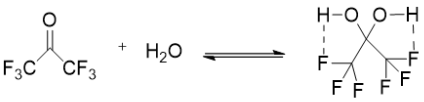
1.

a) 

Baeyer's Strain Theory
The main points of Baeyer's strain theory are:

All cycloalkanes are planar or flat i.e. lie in the same plane. Hence bond angles between an adjacent carbon atom of the ring no longer remain 109° 28′. Different rings have different values of this angle. For example, the cyclopropane ring is an equilateral triangle having a C-C-C angle of 60o only.
Any deviation, positive or negative from the normal tetrahedral bond angle of 109° 28′ during the formation of the ring creates a strain in the molecule which makes the molecule unstable. This deviation of bond angle is known as Baeyer angle strain or simply angle strain.
The larger (more) the deviation from the normal angle, the greater the strain, and thus lesser the stability. However, it should be noted that the sign of deviation does not make any difference.
Mathematically, angle strain = 1/2 (109° 28′ – bond angle in planar form)

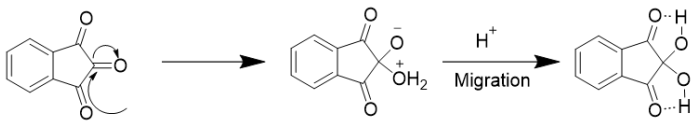
Here the equilibrium will shift in the forward direction due to release of angle strain from sp^2 (120°) to sp^3 (109.5°). In cyclopropanone, the carbon atoms are bonded in such a tight angle of about 60°, which is much less than the ideal angle of 109.5° found in tetrahedral carbons. This significant deviation causes the bonds to be under strain, similar to a tightly coiled spring.Moreover, intramolecular hydrogen bonding can take place. Thus equilibrium shifts in forward direction.



Due to presence of fluorine attached to Hexafluoroacetone (HFA),the electrophilicity of the carbonyl group increases highly. Thus the reaction occurs easily. HFA and acetone adopt very similar structures, the C-O distance being only longer in the fluorinated compound (124.6 vs 121.0 pm), possibly due to steric effects. Hence equilibrium will shift forward.

Here, equilibrium shifts in backward direction because the steric hinderence is high, the epoxide is highly strained, the bond energy for carbonyl group is higher than C single bond O.Thus the reactant is thermodynamically more stable than the product.

2.

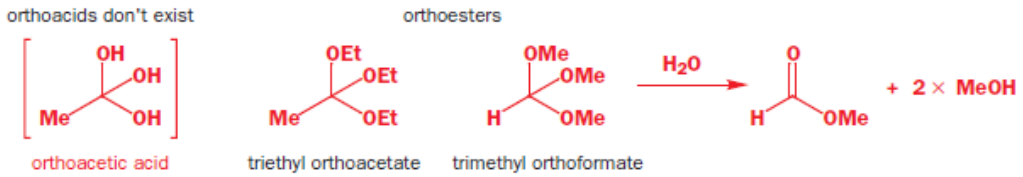


Stabilization by intramolecular Hydrogen bonding.

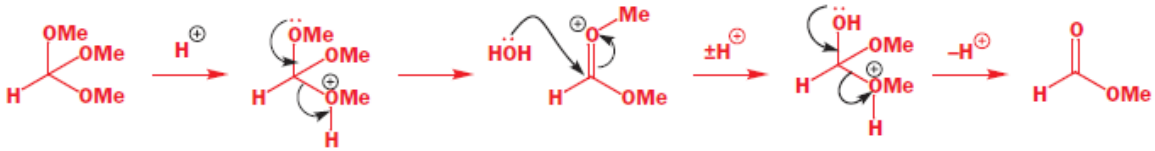
3.

Overcoming entropy: orthoesters

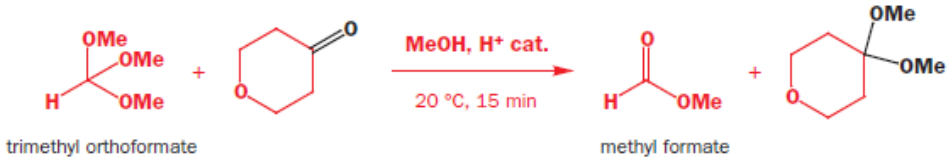
We have already mentioned that one of the factors that makes acyclic *hemi*acetals unstable is the unfavourable decrease in entropy when two molecules of starting material (aldehyde or ketone plus alcohol) become one of product. The same is true for acetal formation, when three molecules of starting material (aldehyde or ketone plus 2 × alcohol) become two of product (acetal plus H_2O). We can improve matters if we tie the two alcohol molecules together in a diol and make a cyclic acetal: we discuss cyclic acetals in the next section. Alternatively, we can use an **orthoester** as a source of alcohol. Orthoesters can be viewed as the 'acetals of esters' or as the triesters of the unknown 'orthoacids'—the hydrates of carboxylic acids. They are hydrolysed by water, catalysed by acid, to ester + 2 × alcohol.



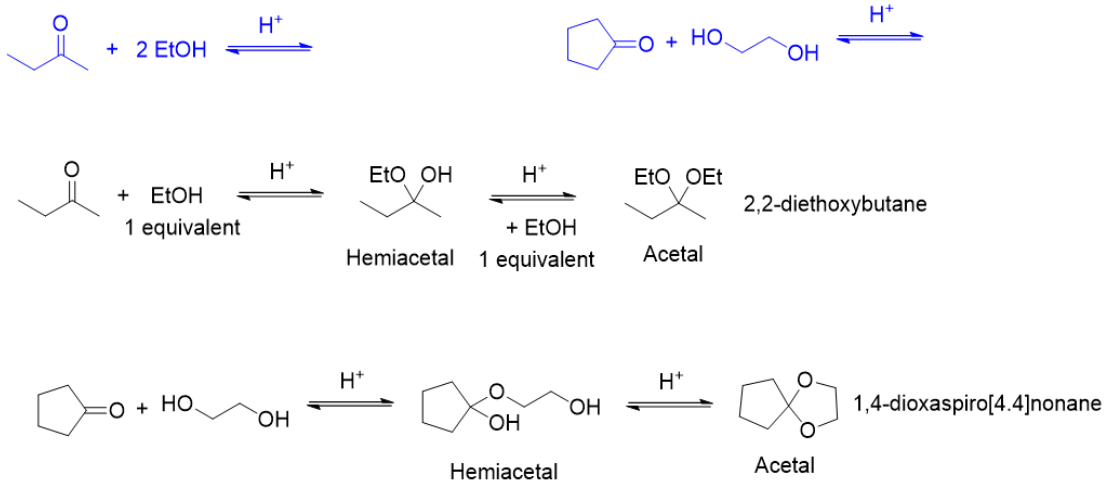
Here is the mechanism for the hydrolysis—you should be feeling quite familiar with this sort of thing by now.



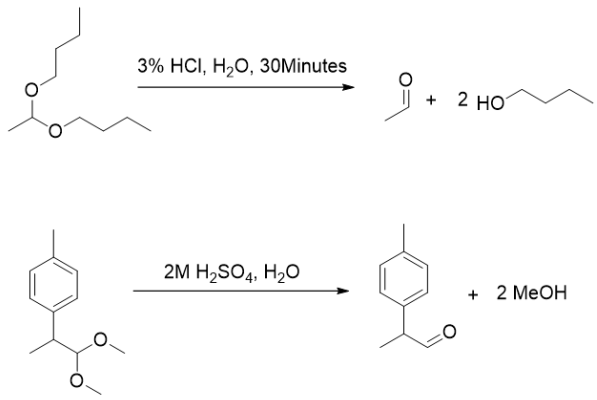
Ketones or aldehydes can undergo **acetal exchange** with orthoesters. The mechanism starts off as if the orthoester is going to hydrolyse but the alcohol released adds to the ketone and acetal formation begins. The water produced is taken out of the equilibrium by hydrolysis of the orthoester.



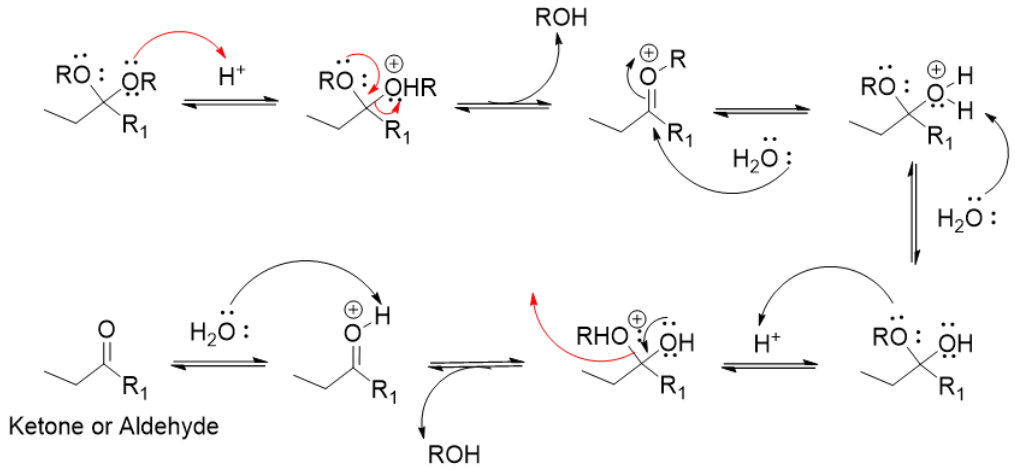
4.



Check notes for the mechanism.



General Mechanism:



5.

See Notes.